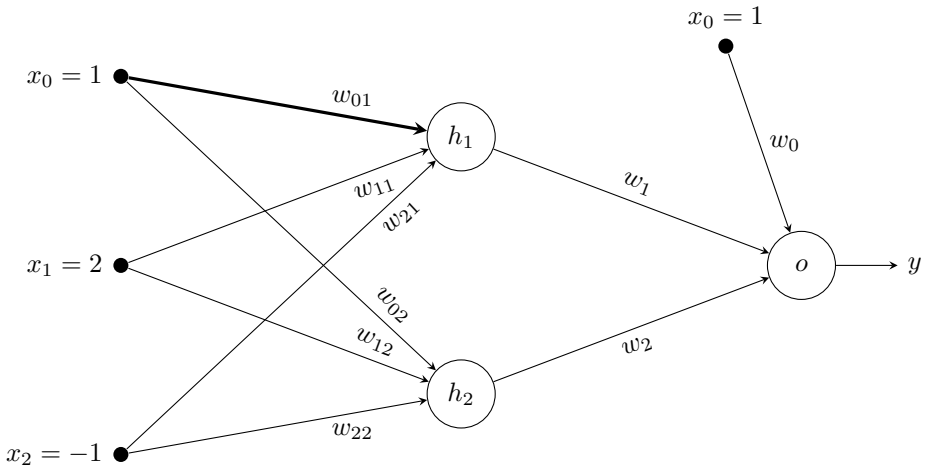
Embedded AI: last practice before the midterm

Four short problems on the three weak points found in the full practice series: the bias input and a hidden weight in backpropagation, the derivative of the sigmoid, and which class is positive in a confusion matrix. Answers on the last page.

1. Neural network with bias inputs, one training step

All neurons use the sigmoid activation. $x_0 = 1$ is the bias input: it feeds both hidden neurons and the output neuron. Input: $x_1 = 2$, $x_2 = -1$. Target: $t = 0.2$. Learning factor: $\eta = 0.2$. Error: $E = \frac{1}{2}(t-y)^2$.



To h_1	$w_{01} = 0.5$	$w_{11} = 0.2$	$w_{21} = 0.4$	To o	$w_0 = -0.4$	$w_1 = 0.7$	$w_2 = -0.5$
To h_2	$w_{02} = -0.3$	$w_{12} = 0.1$	$w_{22} = 0.6$				

a) Forward pass: calculate the outputs of h_1, h_2 and y.
b) Calculate the value of the output weight w_1 after the training step.
c) Calculate the value of the weight w_{01} (bold arrow) after the training step.

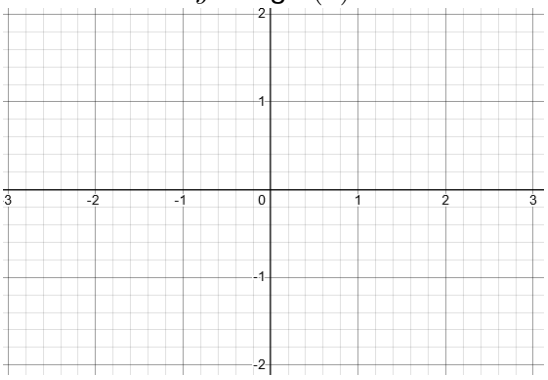
2. Derivative of the sigmoid

a) Complete the table. ($y = \text{sigm}(x) = 1/(1+e^{-x})$, $y' = y(1-y)$.)

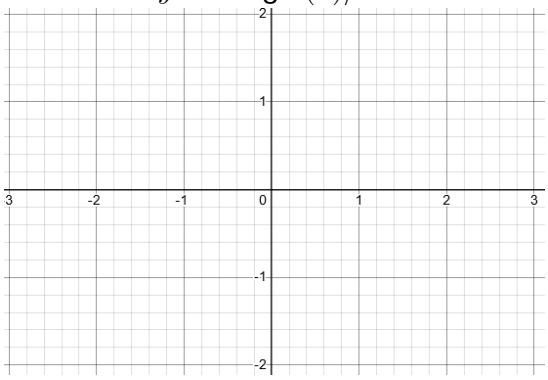
x	-4	-2	0	2	4
$y = \text{sigm}(x)$					
$y' = y(1-y)$					

b) Draw $y = \text{sigm}(x)$ and its derivative y' on the grids.

$y = \text{sigm}(x)$



$y' = \text{d}\,\text{sigm}(x)/\text{d}x$



c) Using the table, explain in two sentences why deep networks with sigmoid activations suffer from the vanishing gradient problem.

3. Confusion matrix

We check our manufactured product. We tested 10,000 products, of which 8,000 are good and the rest are defective. (The defective product is called positive.) We measured a parameter p; the table shows how many good and defective products have p above the limit.

P_{lim}	10	14	18	22
Good products with p above P_{lim}	8,000	7,600	6,200	2,000
Defective products with p above P_{lim}	1,500	600	150	0

a) A product is considered defective (positive) if p is below $P_{\text{lim}} = 14$, good (negative) above. What are TP, TN, FP, FN, the sensitivity and the specificity?

b) Same question with $P_{\text{lim}} = 18$. Which limit would you choose if missing a defective product is very expensive? Why?

Answers

1. a) $net_{h1} = 0.5 + 0.4 - 0.4 = 0.5$, $y_{h1} = 0.622$; $net_{h2} = -0.3 + 0.2 - 0.6 = -0.7$, $y_{h2} = 0.332$; $net_o = -0.4 + 0.7 \cdot 0.622 - 0.5 \cdot 0.332 = -0.130$, $y = 0.468$.
b) $\delta_o = (0.468 - 0.2) \cdot 0.468 \cdot 0.532 = 0.0666$ (positive: output above target, the weights go down). $w_1 = 0.7 - 0.2 \cdot 0.0666 \cdot 0.622 = 0.6917$.
c) $\delta_{h1} = 0.0666 \cdot 0.7 \cdot 0.622 \cdot 0.378 = 0.01095$; $w_{01} = 0.5 - 0.2 \cdot 0.01095 \cdot 1 = 0.4978$. The input of w_{01} is $x_0 = 1$.

2. a) y: 0.018, 0.119, 0.5, 0.881, 0.982. y': 0.018, 0.105, 0.25, 0.105, 0.018.
b) y: S-shaped, from 0 to 1, 0.5 at $x = 0$. y': a bell centred at 0, maximum 0.25, going to 0 on both sides.
c) The derivative is at most 0.25 and close to 0 when the neuron saturates. Backpropagation multiplies one such derivative per layer, so the gradient reaching the first layers shrinks exponentially and they stop learning (fix: ReLU).

3. Defective: $10{,}000 - 8{,}000 = 2{,}000$. The "above" row gives what is classified good (negative).
a) $P_{\text{lim}} = 14$: TP $= 2000 - 600 = 1400$, FN $= 600$, TN $= 7600$, FP $= 8000 - 7600 = 400$. Sensitivity $= 1400/2000 = 0.70$, specificity $= 7600/8000 = 0.95$.
b) $P_{\text{lim}} = 18$: TP $= 1850$, FN $= 150$, TN $= 6200$, FP $= 1800$. Sensitivity $= 0.925$, specificity $= 0.775$. If missing a defect is expensive, choose 18: fewer false negatives (150 instead of 600), at the cost of more good products rejected.